

Ahmad Hossein Yazdani

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Summary

PhD candidate specializing in ML systems infrastructure and performance optimization for large-scale distributed training. Experienced in diagnosing and mitigating GPU, storage, and communication bottlenecks through profiling and systems-level experimentation. Focused on building scalable optimizations for production-scale AI workloads.

Work Experience

August, 2020 – present **Research Assistant at Distributed System and Storage Lab, Virginia Tech.**

Advisor : Dr. Ali Butt, *Professor, Department of Computer Science, Virginia Tech*

- **LLMStore:** Designing caching and activation-ranking mechanisms to reduce tensor offload latency in distributed LLM training; identified 300GB+ (Llama-70B) offload traffic during LLaMA fine-tuning via trace analysis.

- **Silo-based open framework for Federated Learning (2025-):** A **Federated Learning research** project led by Dr. Feiyi Wang (ORNL), Dr. Ali Butt (Virginia Tech), and Dr. Sahil Tyagi (ORNL). Implemented gradient compression in the gRPC communication layer for a cross-cloud/HPC Federated Learning framework, reducing cross-site communication overhead and enabling hierarchical aggregation.

- **Metis** The project introduces a data access pattern for Deep Learning workloads to improve cachability. With a small cache, it improves the hit ratio by up to $4.5\times$ compared to the LRU policy : **Shade paper at FAST23**

January, 2026 – **Research Intern, Oak Ridge National Laboratory.**

Mentors: William Godoy, Pedro Vallerio

- Selected for ORNL research internship (2026) to explore AI-driven optimization of memory- and storage-intensive HPC workflows.

June, 2024 – August, 2024 **Student Assistant at NERSC, Lawrence Berkeley National Laboratory (LBNL), internship.**

Mentors: Stephen Simms, Lisa Gerhardt, Jean Luca Bez

- Examined Drishti, an HPC I/O recommendation tool; Detected many false positive warnings, and derived insights to improve the accuracy of the the Drishti I/O recommendation tool.

June, 2023 – August, 2023 **Student Assistant at Lawrence Berkeley National Laboratory (LBNL), internship.**

Mentors: Suren Byna, Jean Luca Bez

- Identified a significant I/O variability for various HPC applications like E3SM (For earth modeling) and LAMMPP (For molecular simulations) when other applications interfere with it. Later continued my research for Large AI models and language models : **LLMStore**

June, 2021 – August, 2021 **Internship at Oak Ridge National Laboratory, Analytics & AI Methods at Scale Group.**

Mentors: Feiyi Wang, Sarp Oral, Ahmad Maroof Karimi and Arnab Kamur Paul

- **This experience resulted in my ICDCN'25 paper, where I was able to predict the I/O pattern of the next job given the features from the past submissions of the same user with an accuracy of nearly 90% for HPC jobs.**

Computer skills

Programming Languages & frameworks	Python, R, C, C++, Advanced JAVA EE, JAVA SE, Tensorflow, Go, Rust, MPI, pthread, OpenMP, CUDA, Java bytecode, X86 assembly
ML	PyTorch, keras, scikit-learn, DeepSpeed, LLM, Computer Vision, Federated Learning
Systems	Distributed systems, File Systems, Lustre, GPFS, Slurm, MapReduce, Redis, Ceph, NVMe, Darshan, Docker

Conference & Workshop publications

- [ICDCN'25] **Yazdani, Ahmad Hossein**, Arnab K. Paul, Ahmad Maroof Karimi, Feiyi Wang, and Ali Butt. User-based i/o profiling for leadership scale hpc workloads. In *Proceedings of the 26th International Conference on Distributed Computing and Networking*, ICDCN '25, page 181–190, New York, NY, USA, Jan. 2025. Association for Computing Machinery. doi:10.1145/3700838.3700865.
- [FAST'23★] Redwan Ibne Seraj Khan, **Yazdani, Ahmad Hossein**, Yuqi Fu, Arnab K Paul, Bo Ji, Xun Jian, Yue Cheng, and Ali R Butt. Shade: Enable fundamental cacheability for distributed deep learning training. In *Proceedings of the 21th USENIX Conference on File and Storage Technologies*, page 14, Santa Clara, CA, US, Feb. 2023. USENIX Association. URL <https://www.usenix.org/conference/fast23/presentation/khan>.

★ Top-tier venue

Education

- 2020–August 2026 **PhD, Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt
- May 2025 **Masters of Computer Science**, *Virginia Polytechnic Institute and State University (Virginia Tech)*, Blacksburg, VA, US.
Advisor: Dr Ali Butt
- 2015–2020 : **Bachelor of Computer Software Engineering**, *University of Tehran*, Tehran, Iran.